PAPER_1310_HIGGS_TRILINEAR

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PAPER_1310 — Higgs Trilinear Coupling λ_HHH

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: C — Particle Physics Open (CLOSED)
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Calculator surface: `calculate_paradox({"paradox": "higgs_trilinear"})`
Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1310})`

Master Expression

``
`κ_λ = λ_HHH/λ_SM = 1.0 (SM-like, no anomaly)`
``

Match: STRUCTURAL

Interpretation

F_U = 1 closure prevents anomalous Higgs self-coupling. Testable HL-LHC κ_λ ∈ [0.1, 2.3].

UQFF Locked Primitives

- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, Λ = 0.00729735
- Λ_QCD = 0.217 GeV, v_Higgs = 246 GeV (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.
